



New Hogwarts Battle

Hogwarts battle is a new tabletop deck-building Harry Potter game where you try and beat Voldemort. <http://dgit.in/HogwartsB>



600 million Instagramers

Facebook's photo and video sharing app Instagram now has over 600 million active users. <http://dgit.in/InstagramF>

HOW WE TESTED

Testing rig and setup: To maintain a standard workflow, we've tested all the mice on a single rig since we discovered how the testing software could suffer anomalies with certain settings. The software that we used to test the accuracy of the sensors were highly sensitive to certain programs running in the OS which had to be disabled while some had to be enabled. Hyperthreading had to be enabled (usually enabled by default), HPET mode had to be disabled, USB Selective Suspend had to be disabled, essentially, anything interfering with the state of the CPU had to be disabled to ensure they didn't utilise the CPU while recording the sensor data. Only the mouse being tested and a keyboard was connected to the rig while keeping the remaining USB ports free to avoid any error.

Although a robotic arm clamping the mouse over a spinning turntable would have been the ideal setup, instead we decided on restricting the window of collecting the sensor data. This step ensured that the data gathered from the different mice fell under the same constraints and it becomes easier to calculate a final score. We used four different type of surfaces to test including a cloth mouse pad, the Razer Goliathus Control Edition mouse pad, a white hard plastic desk and a glass slab with a black

cloth underneath. Measurements were marked on the desk itself to avoid using a ruler every time, which would have obviously introduced errors every time. Around this area, a rough enclosure was made to restrict the movement of the hand to execute accurate motions.

Performance test: To introduce objectivity, we used several software. The most important among them are MouseTester v1.5.3 and Enotus Mouse Test v0.1.4. MouseTester has built-in plotting of sensor data that lets you generate graphs depicting many parameters such as accuracy, polling rate, velocity, etc. We used MouseTester for every mouse set at 800 CPI to obtain the sensor's accuracy when the mouse was given a quick swipe on a cloth mouse pad. Overall, only three CPI levels were used for all the tests – 800, 1,600 and 3,200. To maintain consistency, we ensured the swipe speed between a velocity range of 2-2.4 m/s, keeping the total time under 300 ms. Mouse settings had to be verified in the OS (Windows 10) as well, so the Pointer Speed was kept at the default 6/11 with Enhance pointer precision disabled (since it introduces acceleration).

Enotus Mouse Test was used to obtain certain parameters, maximum speed being an important one. The maximum speed at which the mouse could detect movement was

recorded, a test limited by the reviewer's ability to swipe the mouse as fast as possible. This brings in a real world parameter into the score and gives us information on how differently a mouse behaves on different surfaces.

In the beginning, the CPI was calculated using the Measure tool in MouseTester and to bring in more accuracy, it was calculated in MS Paint also (moving the cursor to the left edge of the canvas on x-coordinate 0 and then moving the mouse one inch using a ruler, gives you the DPI from the value of the x-coordinate). The CPI was calculated every time prior to running any benchmark that involved an objective test, and also to verify whether different DPI levels advertised by the manufacturer were accurate.

Measuring the lift-off distance was done using the age old DVD technique where the mouse was kept on one DVD and then more added until the sensor stopped detecting. DVDs have a standard thickness of 1.2 mm, so if the sensor detected over one DVD and stopped detecting with the second DVD, the recorded lift-off distance was 1.2mm. Lift-off distance customisation was switched off in the software (if available).

Response time was evaluated but it was a subjective test since it was dependent on the reviewer's reaction time. This also gives us a

best looking RGB mouse. The build quality is amazing with rubber coating, textured at the finger grips, held together with a metal frame inside, and giving you options of replaceable side skirts.

Logitech Proteus Core G502

The most popular mouse in this comparison and a flag bearer in overall accuracy, comfort, build quality, features and productivity, the G502 is a dream gaming mouse for many. Continuing the legacy of the G-series, the G502's main attraction is the exclusive sensor – the legendary 3366. But it comes at a price and weight, the latter being stressed because it's comparatively heavier at

around 120 g. The infinite scroll wheel is a wonderful feature that we found ourselves using. Currently, the Spectrum version (RGB) is available as well, selling at around the same cost. You know what to do.

Razer Death Adder Elite

Razer continues its tradition of improving the venerable Death Adder every year, this time in the form of Elite. It features the same iconic design with better texture on the



Razer DeathAdder Elite

side buttons, a better sensor which is highly accurate, and finally custom manufactured switches with Omron with a durability of 50 million clicks. The mouse is perfect for claw grip and the wide feet ensure the necessary amount of friction is obtained. Our only qualm is with the limited features in the software and the slight jitter at 1,000Hz. Other than that, it's close to a perfect mouse, and only time will tell about its durability.

SteelSeries Rival 100 and 300

RGB lighting is present in almost every mice in this comparison, and other than indicating DPI, none of them have necessary functionality. SteelSeries has the best imple-

mentation with GameSense where your actions and state in a game affects the lights on your mouse. This feature works well and we tried it out in CS: GO. Speaking about the mice, the 100 is a great budget option for a small gaming mouse with limited control to the DPI settings in the software whereas the 300 is a better option, meant for bigger hands and has a better sensor. They are available in different colours and accents.

Tt eSports Ventus Z

The Ventus Z brings back the meshed design we had seen on the Level 10 M Advanced RGB. Although, the mesh didn't offer any functionality, other than just aesthetics, in the latter because